

Augmented Offense — Module 3:

H@nd\$hak3 (T5577 125 kHz LF)

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~19 min (1:04–1:23) · Flipper or Proxmark — no phone

Module 3 — H@nd\$hak3 (T5577 125 kHz LF), for complete beginners

You have never used a Proxmark or Flipper? Perfect — this assumes that. Follow it top to bottom. Do exactly what each numbered step says and check the green “✅ **You should see**” box before moving on. The “⚠️ **If not**” box tells you what to do when something’s wrong.

Implant analog: xEM / flexEM. **Time:** ~19 minutes.

No phone can do this module. 125 kHz has no radio in any phone. You need a **Flipper Zero** (works on its own) or a **Proxmark3** (needs a laptop). We can hand you a Flipper if you don’t have a reader.

1. What are you about to do, in plain English?

The old **125 kHz “prox” fob** is the round keyfob or thin white card that opens a lot of doors. It has **no password** — it just shouts a number, and the door believes it. That’s the entire security model, and it’s broken.

At the front of the room there is a **master puck** (213’s own retired door-strike key). You are going to **read its number and copy it onto your own blank card**. When you’re done, your card opens whatever the master opens — *“you all now carry the master key.”* That’s the point: copying a prox card is trivial and needs no secret.

You are copying **213’s provided master**, on **your own blank card**. Never copy a real, in-use badge.

2. Mini-glossary

- **EM4100** — the most common 125 kHz prox format. Its number is 10 hex characters (0–9, A–F).
 - **T5577** — a blank card you can write any prox number onto. That's your target.
 - **Master puck** — the physical card at the front you'll copy from.
 - **Reader** — your **Flipper Zero** (handheld, standalone) or **Proxmark3** (board on a USB cable, needs a laptop).
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3. What you need at your seat

- ☐ A **Flipper Zero** or a **Proxmark3 + laptop** (ask a pod lead if you need a Flipper)
 - ☐ Your blank **T5577 card**
 - ☐ Brief access to the **master puck** (shared at the front / passed around by a pod lead)
 - ☐ The pod **demo reader** to test your finished card
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4. Pick your path — DO ONLY ONE

This module has two paths. You do **ONE** of them, based on the reader in your hand. They are alternatives, **NOT** steps — do not do both.

- Holding a **Flipper Zero**? → do **Path A**, then stop. (*Skip Path B entirely.*)
- Holding a **Proxmark3 + laptop**? → **skip Path A** and do **Path B**.

Not sure which you have? The Flipper is a small handheld with a screen and a dolphin. The Proxmark is a bare circuit board on a USB cable. Ask a pod lead.

Path A — Flipper Zero (*skip this whole section if you're using a Proxmark*)

The Flipper is a handheld with a screen and arrow buttons. The **125 kHz antenna runs around the edges** — the master puck and your T5577 sit **flat on the back** of the Flipper.

Buttons: center round button = **OK/Select**. Arrows move. **Back** = previous screen.

Menu wording varies slightly by firmware. If a label isn't exact, pick the closest match — the flow is the same.

A1. Open the 125 kHz app

1. Press **OK** for the main menu.
2. Arrow to **125 kHz RFID** and press **OK**.

A2. Read the master puck

1. Choose **Read**.
2. Hold the **master puck flat against the back** of the Flipper and keep it still.

✅ **You should see** a card read, showing a format like **EM4100** and a number.

⚠️ **If not:** “Reading...” forever → the puck isn't positioned right. Lay it flat centered on the back, don't move it, and wait. Only one card on the reader at a time.

A3. Save it

1. Choose **Save**, name it `master`, confirm.

A4. Write it onto your blank T5577

1. From **125 kHz RFID** → **Saved**, open `master`.
2. Choose **Write**.
3. Put your **blank T5577 flat on the back** of the Flipper and hold still.

✅ **You should see** “Successfully written” (or similar).

⚠️ **If not:** “Write failed” → the T5577 slipped. Re-center it on the back and try again. Make sure it's the **blank T5577**, not the master.

A5. Prove it worked

1. From **125 kHz RFID**, choose **Read**.
2. Hold your **T5577** to the back.

✅ **You should see** the **same number as the master**. Your card is now a copy. Tap it on the pod **demo reader** to confirm it reacts like the master. 🎉

🛑 **STOP — you're done with Module 3.** Do NOT continue into Path B below; that's the Proxmark version of the exact same thing. Move on to Module 4.

Path B — Proxmark3 *(the alternative to Path A — skip this if you already did Path A on a Flipper)*

The Proxmark3 is a small board on a USB cable. The flat **round coil is the LF (125 kHz) antenna** — cards go there. Everything below is typed at the Proxmark console.

B1. Open the console

1. Plug the Proxmark3 into the laptop's USB.
2. In a terminal, type `pm3` and press **Enter**. (If “not found,” try `./pm3` from the Proxmark folder, or ask a pod lead.)

✅ **You should see** a prompt:

```
[usb] pm3 -->
```

`[usb]` means it's connected. Type every command below at this prompt.

⚠️ **If not:** `[offline]` → not connected. Replug the USB, try another port, run `pm3` again.

B2. Read the master puck (get its number)

1. Lay the **master puck flat on the round LF antenna**.
2. Type: `lf search` and press Enter.

✅ **You should see** it identify the card, e.g. `EM410x ID found` with a 10-hex number like `1A2B3C4D5E`. **Write that number down** — it's what you clone.

⚠️ **If not:** “No known 125 kHz tags found” → re-center the puck on the coil and run `lf search` again.

B3. Write it onto your blank T5577

1. Take the master off, lay your **blank T5577** on the antenna.
2. Type, replacing `1A2B3C4D5E` with the number from B2:

```
lf em 410x clone --id 1A2B3C4D5E
```

✅ **You should see** a confirmation that it wrote the EM4100 ID to the T5577.

(If B2 reported HID instead of EM4100, use `lf hid clone` with the values it showed — ask a pod lead; most master pucks are EM4100.)

B4. Prove it worked

1. Keep the T5577 on the antenna. Type: `lf search`

✓ **You should see** the **same number** you wrote. That's the clone confirmed. Tap it on the pod **demo reader**.

🛑 **STOP — you're done with Module 3.** (Path A was the Flipper version of this same thing — you didn't need it.) Move on to Module 4.

5. How do I know I succeeded?

Whichever **one** path you did ends the same way: **read your T5577 and see the master's number**, then **the pod demo reader reacts to your card the same as the master**. If both are true, you cloned the master.

6. Why it matters

You just copied a door key by holding it near a \$30 gadget — no password, no hack, no trace. Every low-frequency prox badge on Earth works exactly like this. An implant (like an xEM) *is* that copy: no card to carry — the operator walks up and they're the key.

Defensive angle: physical access control that trusts a broadcast number has no defense here — the reader can't tell a clone from the original, and it can't tell a card from a hand. Retire 125 kHz prox.

7. Troubleshooting

What you see	What it means	What to do
“Reading...” never finishes	card not positioned	lay it flat/centered (Flipper back / Proxmark round coil), hold still, one card only
Proxmark: “No known 125 kHz tags”	reader can’t see it	re-center on the coil, retry <code>lf search</code>
Flipper: “Write failed”	T5577 slipped	re-center on the back, retry; confirm it’s the blank
Clone reads a different number	wrong ID typed (Proxmark)	re-check the number from <code>lf search</code> , redo the clone
Proxmark prompt <code>[offline]</code>	not connected	replug USB, other port, run <code>pm3</code> again
Everyone’s card is identical	expected — you all copied one master	that IS the lesson: “you all carry the master key”

8. Safety (non-negotiable)

- Copy **only** the provided master puck, onto **your own** blank T5577.
- **Never** read or clone anyone’s real building badge, keyfob, or hotel/transit card — not even “just to see.”
- These skills are illegal against systems you don’t own or aren’t authorized to test. Here, on the provided master and your own blank, you’re clear.

Presenters

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